

BIBHU MISHRA

Principal Architect | Cloud, DevOps & Generative AI | Enterprise Delivery Leadership

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EXECUTIVE SUMMARY

Principal-level technology leader with 18+ years spanning hands-on engineering, enterprise solution architecture, and delivery leadership of onshore/offshore teams up to 50 — currently shipping production GenAI solutions (agentic RAG, fine-tuned LLMs, MCP-based tooling) at Microsoft. Proven record of turning ambiguous business goals into secure, scalable cloud-native and AI platforms, then guiding the teams that build, ship, and operate them. Trusted advisor to enterprise customers including Microsoft and Tenet Healthcare — owning roadmaps, managing expectations and escalations, and reporting to executive stakeholders. Deep domain grounding in Healthcare (EHR, HL7, HIPAA) and Content Management, paired with current expertise in Azure, Kubernetes, Terraform, and LLMOps. Equally effective defining architecture, mentoring engineers, and owning multi-million-dollar delivery commitments.

CORE COMPETENCIES

Leadership & Delivery: Team leadership & mentoring (teams up to 50) · Program & delivery management · Customer & stakeholder management · Pre-sales / RFP–RFI response · Governance & delivery excellence · Agile & Scrum

Architecture & Cloud: Enterprise & solution architecture · Azure (AKS, App Services, Functions, API Management, Front Door, Cosmos DB) · Kubernetes & containers · Terraform / IaC · Microservices

Generative AI & LLMOps: Agentic AI & RAG architecture · Open AI SDK / LangChain / LangGraph / AutoGen · LLM fine-tuning · MLflow / Langfuse observability · Azure OpenAI · Azure AI Search · Microsoft NLWeb · MCP (Model Context Protocol)

DevSecOps & Reliability: CI/CD (Azure DevOps, GitHub) SAST / DAST / SCA (SonarQube, CodeQL, Snyk) Zero Trust / Microsoft SFI · Prometheus, Grafana, Azure Monitor · SRE

Programming & Scripting: Strong hands-on experience with Node.js, Python, Vue.js, NestJS, FastAPI, FastMCP, and shell scripting, applied to building scalable applications, backend services, automation workflows, and CI/CD pipelines.

Domain Expertise: Healthcare (EHR, HL7, HIPAA, EMR) Content Management & Digital Marketing · Capital Markets · Utility (AMI / MDM) Credit (PLCC)

PROFESSIONAL EXPERIENCE

Microsoft (via LTI Mindtree)

Dallas, TX

Principal Architect — DevOps, Microsoft Marketing Web (GDC)

Sep 2022 – Present

Lead architect and delivery owner for end-to-end DevOps and SRE across 100+ Microsoft marketing websites — infrastructure provisioning, CI/CD, application security, deployments, and continuous monitoring — accountable for a secure, reliable, high-performing digital experience for Microsoft's end customers.

- **Led the DevOps/SRE team end-to-end** for Microsoft's GDC Web organization, setting technical direction and mentoring engineers while ensuring high availability, reliability, and operational excellence across the marketing web estate.
- **Architected and migrated 40+ Microsoft.com blog and news sites to AKS**, reducing infrastructure cost by 40%; built self-healing, auto-scaling clusters with Terraform and Helm, and secured workloads with RBAC, pod security policies, container image scanning, and Azure AD authentication.
- **Modernized AKS ingress by migrating from the NGINX Ingress Controller to Azure Application Gateway Ingress Controller (AGIC)**, evaluating Application Gateway for Containers (AGC) against AGIC across feature readiness, compatibility, security, and operational requirements, and making the architecture decision to adopt AGIC for the current production landscape.
- **Re-architected search across Microsoft marketing websites**, replacing self-managed ElasticPress with Azure AI Search — designed the content ingestion and vectorization (embedding) pipeline and implemented hybrid retrieval combining keyword, vector, and semantic ranking; materially improved search relevance and content discoverability while eliminating Elastic infrastructure maintenance overhead.
- **Engineered reusable Azure DevOps CI/CD pipelines and templates** integrating SAST (CodeQL), Microsoft Defender, and automated testing; automated infrastructure deployment with Terraform, reducing release errors and improving deployment efficiency across environments.
- **Embedded security, privacy, and resilience into application design and delivery** by partnering with project development teams, Microsoft Security, and third-party vendors to create and review threat models, complete privacy reviews and SDL assessments, establish BCDR readiness, and align external delivery teams with Microsoft security standards and controls for a consistent, secure, and efficient SDLC; implemented continuous security validation through SAST with CodeQL, dependency and supply-chain analysis with Dependabot and Microsoft Defender, and DAST using Microsoft proprietary tools including WCP and URSA, with vulnerability triage and remediation aligned to MSRC processes.
- **Provided operational governance and leadership across incident, service request, security, and release management**, establishing clear ownership, prioritization, escalation, and execution practices; regularly monitored Microsoft S360 for security findings, coordinated remediation with engineering teams, and drove issues through timely closure.
- **Automated day-to-day SRE operations** with custom tooling built on Azure Automation, Python, LangChain, and Azure OpenAI, and built centralized monitoring dashboards with Datadog, Prometheus, Grafana, and Azure Monitor.

Principal Architect — GenAI Enablement Practice

Feb 2026 – Present

Founding architect of a GenAI enablement practice within Microsoft's Business Unit, chartered to operationalize generative AI across development, support, and DevOps organizations.

- **Shaped the practice's architecture, tooling strategy, and adoption roadmap**, partnering with product and engineering leaders to prioritize high-impact use cases and deliver measurable productivity gains.
- **Delivered a portfolio of production RAG solutions: SRE Buddy** — an agentic knowledge RAG over SRE operational SOPs, integrating ICM, Kusto, and Azure DevOps to accelerate incident response; **AICA (AI Content Analyzer)** — mines user comments from Microsoft blog sites and chatbot interactions on marketing websites, distilling sentiment and user-experience insight into actionable reports for management; and **Microsoft NLWeb** — implemented to bring conversational, natural-language search to marketing web properties, improving user experience and engagement.
- **Implemented Azure SRE Agent** across the incident-management workflow — automating ticket classification, intelligent routing to the correct resolver groups, and first-line diagnostics (log retrieval, health checks, known-error matching) before an engineer engages. Cut manual triage effort significantly; now extending the agent with runbook automation to auto-resolve recurring incident classes and further reduce MTTR.
- **Incubating LLM fine-tuning for Cause Code Analysis** — automating root-cause categorization of incidents; projected to save one full-time resource of manual classification effort while improving categorization consistency and accuracy.
- **Built AKS cluster investigation tooling** with the Grafana suite and a Grafana MCP-powered chatbot — engineers query cluster health, metrics, and dashboards in natural language, shortening diagnosis time during SRE triage.
- **Built a consolidated status-reporting dashboard** giving leadership a single-pane view of incidents, DevOps user stories, and S360 KPIs via Microsoft MCP servers — streamlining executive status reporting and team tracking, extending it with Grafana and Azure MCP servers to fold infrastructure and application monitoring into the same view.
- **Established a scalable AKS hosting and CI/CD model** with Langfuse and Grafana observability for prompt/response evaluation, governance, and guardrails.
- **Drove GenAI enablement and responsible-use standards**, coordinating training with vendors and SMEs to upskill multiple engineering teams.

Tenet Healthcare (via NTT Data / Dell)

Dallas, TX

Technical Architect & Delivery Lead — DevOps & App Modernization

Nov 2021 – Sep 2022

Modernized and standardized Tenet's application delivery platform through enterprise-wide Azure DevOps adoption — migrating legacy TFS, strengthening security and compliance, and establishing scalable CI/CD across diverse technology stacks to improve developer productivity, release reliability, and visibility.

- **Spearheaded enterprise Azure DevOps adoption**, migrating 60+ projects from legacy TFS — streamlining version control (Git), work-item tracking (Azure Boards), and reporting (Azure Monitor, Power BI) across the account.
- **Developed and optimized YAML-based CI/CD pipelines** for .NET, Node.js, Java, and Python applications, integrating AKS, Helm, and Docker containerization; instituted a Git-flow branching strategy with cross-functional teams to modernize the development process.
- **Designed and enforced security controls** — Azure AD, RBAC, service connections, pipeline approvals, and branch policies — and integrated SonarQube code scanning into pipelines to enforce Secure SDLC compliance.
- **Led migration of 48 consumer websites to a custom ASP.NET Core / Vue.js / Sitefinity CMS**, raising availability to 99.99% and Lighthouse scores from single digits to 80%.
- **Ensured a seamless, low-risk transition** through hands-on technical guidance and troubleshooting, and conducted training and workshops on Azure DevOps best practices, security automation, and cloud-native deployment strategies.

Tower Leader — Enterprise App Support & SharePoint Modernization

May 2015 – Nov 2021

Owned a \$5M/year enterprise portfolio (consumer sites, SharePoint portals, BPM apps) serving 160K users across 48 hospitals; led onshore/offshore teams of 20+.

- **Directed incident and problem management under ITIL**, reducing ticket volume 60% over two years through automation and root-cause analysis.
- **Managed customer feedback, escalations, and executive stakeholder communications**, ensuring SLA compliance and alignment with business priorities.
- **Transitioned applications from third-party vendors to in-house support**, reducing operational cost and improving service continuity.
- **Led consolidation and migration of a 4TB+ SharePoint estate** (40+ hospital intranets, 40+ consumer sites) with governance, access control, and end-user training.

Dell Services

Bengaluru, India

Practice Delivery Leader — Microsoft SharePoint Practice

Apr 2013 – Apr 2015

Built and led Dell's SharePoint practice within the Microsoft horizontal, managing a 25-person team across Bengaluru and Noida.

- **Institutionalized delivery excellence** — reusable asset repository, coding standards, governance, and operational reviews.
- **Developed fixed-price service offerings** (SharePoint migration, knowledge management, ITIL) with sales collateral, supporting pre-sales and RFP/RFI response.
- **Led capability building** through training, certification, and demand-supply planning; partnered with Dell Software Group on optimized delivery models.

Project Lead — BizTalk Orchestration (Owens & Minor)

Nov 2011 – Mar 2013

- **Led offshore delivery of a centralized BizTalk ESB layer** replacing point-to-point integrations; owned WBS, estimates, quality reviews, audits, and stakeholder reporting.

Team Lead — Microsoft Amalga EMR Program

Sep 2009 – Nov 2011

- **Built Amalga UIS competency for Dell and led implementations** at Moses Cone, Norton Healthcare, and St. Joe Hospitals using HL7/XML standards, managed offshore delivery and quality.

EDUCATION & CERTIFICATIONS

- **Post Graduate Diploma in IT (Executive Management)**, VGSOM, IIT Kharagpur, India
- **B. Tech, Electrical Engineering**, CET, OUAT Bhubaneswar, India
- **Postgraduate in AI/ML & GenAI** — In Progress

Certifications: AZ-400 (DevOps Engineer Expert) AZ-104 (Azure Administrator) CKAD · Terraform Associate · PMP · PSM I (Scrum Master) HL7 Healthcare Messaging